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Data Communication & Computer Network (DCN)

Unit 1 + Unit 2 Notes

Data = Information presented in waveform

→ Data Commⁿ:

↓

Exchange of Data Between two Devices via Same medᵐ

→ 4 Fundamental Char::

a) Delivery b) Accuracy c) Timeliness d) Jitter (Variation in packet arrival time).

Data Representation::

→ Text
→ Image
→ Audio / Video

Data Flow::

1) Simplex ⇒ Communication is Unidirectional i.e TV

⇐ Unidirectional

2) Half Duplex ⇒ Either direction but only one way at a time i.e Police radio (walkie-talkie)

⇆

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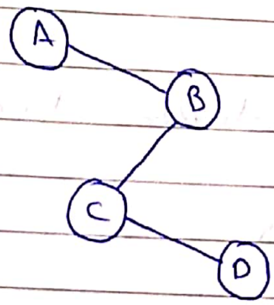
3) Full Duplex => Both Direction at Same time.
i.e Phone

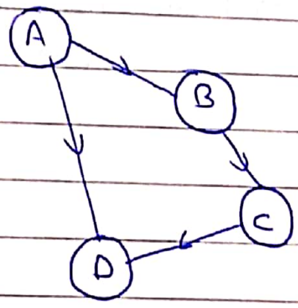
$\longleftrightarrow$

Connection :-

1) Point to Point -



2) Multipoint -



Medium :-

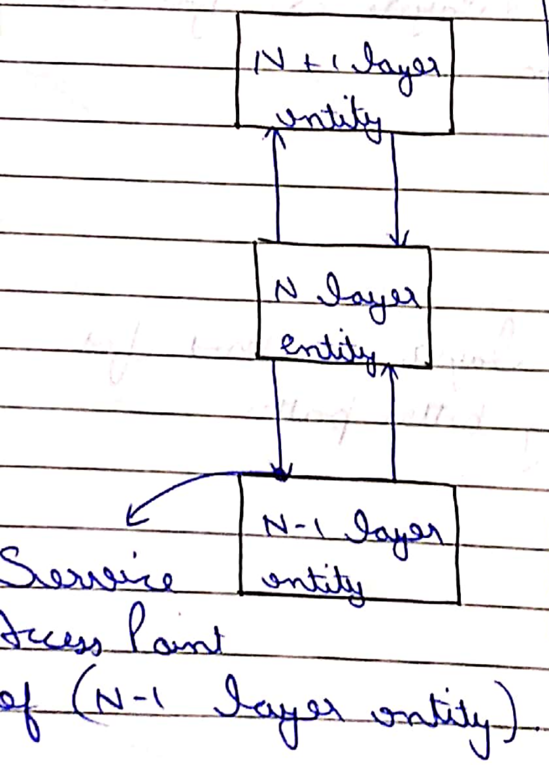
1) Guided = Wire / Cable.

2) Unguided = Wireless.

Note :-

In Data Communication, Communication is in the form of electromagnetic waves.

Layered Network Architecture	Protocal Architecture
→ In this Architecture Services are grouped in a hierarchy of layers.	→ Protocal Architecture uses two types of model.
→ An entity of layer N uses only Services of layer N-1	→ OSI reference model → 7 layers (Protocal)
→ An entity of layer N provides only Services of layer N+1	→ TCP / IP → 4 layers (Protocal)



OSI Model

Application
Presentation
Session
Transport
Network
Data link
Physical

TCP / IP

Application	→ FTP, SMTP, DNS
Transport	→ TCP, UDP
Internet	→ ARP, IP, ICMP, IGMP
Network Interface layer	→ Ethernet, Token ring, ATM

OSI - Open System Inter Connection Model

↳ It is conceptual model that describes standard for inter Computer - communication.

↳ Each layer is connected with the Second layer in the go to provide Service

↳ OSI model has 7- layers for the Communication

- Application
  - Interface ← ↑ Services
- Presentation
- Session
- Transport

These 4 layers are for end System

- Network
- Data link
- Physical

These 3 layer are Used for providing better paths

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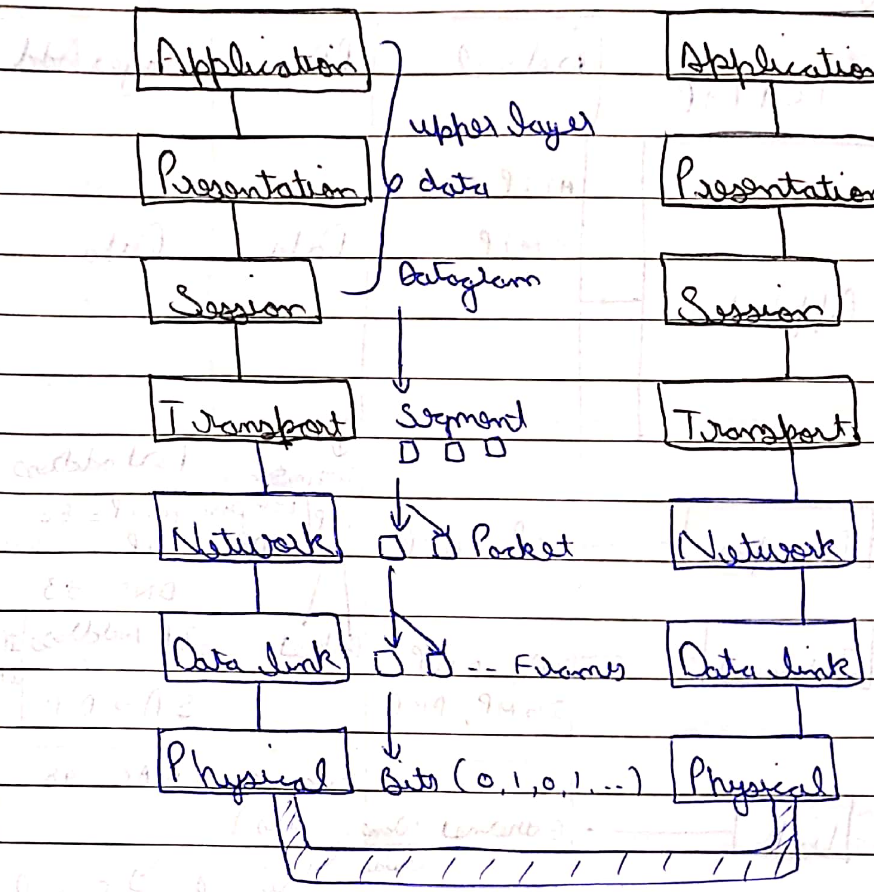

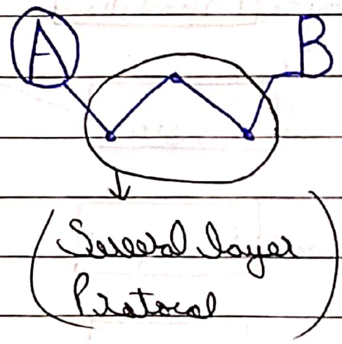

Please Do Not Tell Secret Password Anyone

Application = File Transfer, e-mail, remote login.

Presentation = Formatting, Compression, Encryption / Decryption.

Session = Establishes, manages & terminates $Conn^n$, Control dialog

Transport = Provides end-to-end reliable delivery (Flow control)

Network = Addressing and routing.

Data link = Provides $Conn^n$ b/w hosts on Same n/w (Ethernet, mac address)

Physical ~~Presentation~~ = Describes electrical & physical $Spec^n$ for devices Cable Connect

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TCP/IP Model

Devices used	OSI	TCP/IP	Protocol	PDU	Layer Add
Gateway	Application	Application	HTTP, FTP, SMTP	Data	Data
Gateway	Presentation	Application			
Gateway	Session	Application			
Firewall	Transport	Transport	TCP UDP	Segment (TCP) / Datagram (UDP)	Port address: HTTP = 80, FTP = 20 & 21, DNS = 53
Router	Network	Internet	IP, ICMP, IGMP, ARP	Packet	IP Address (IP4 / IP6); S.A → D.A
Bridge	Data link	Link	Ethernet, Frame relay	Frame (bits) → Signal	MAC - 48 → Signal
Repeater	Physical	Link			

Note :

HTTP = Hyper Text Transfer Protocol
FTP = File Transfer Protocol
SMTP = Simple Mail Transfer Protocol

S.A = Source Address
D.A = Destination Address

TCP = Transport Control Protocol.
UDP = User Datagram Protocol.

ICMP = Internet Control Message Protocol.
IGMP = Internet Group management Protocol.
ARP = Address resolution Protocol.

PDU = Protocol Data Unit

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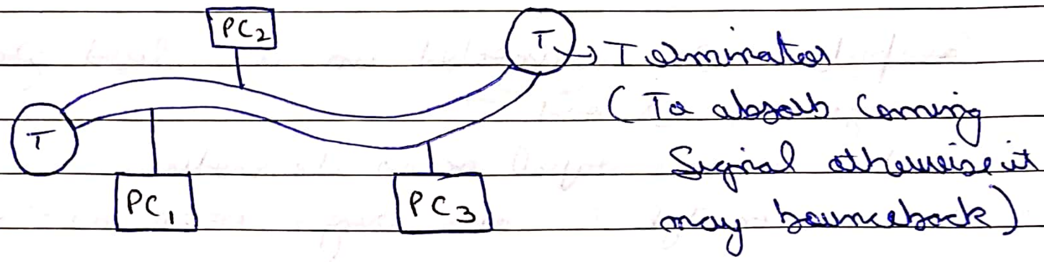
→ Topology - It is the Arrangement with which Computer and network devices are connected to each other.

→ Arrangement layout can be actual and logical.

Bus :



→ Uses trunk or backbone to which all of the computers on the n/w are connected.

→ Cable Seg must end with a terminator

→ Thick ethernet (10 Base 5) used for backbone.

Advantage	Disadvantage
→ cheap & easy to implement	N/W disruption when Comp are added or removed
→ Require less cable.	A break in backbone leads to whole System down
→ Work well for Small network.	Sharing Same Cable → Slow response.

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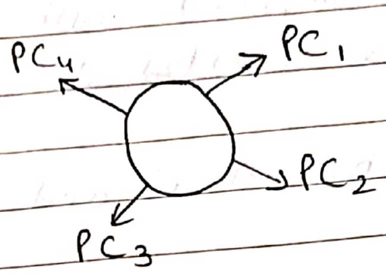
Ring :



- Computers are Connected in a Closed group.
- No beginning or end.
- All devices have equal access to media.
- Most common type (Token ring = IEEE 802.5)

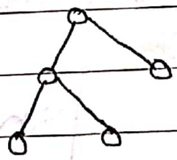
Advantage	Disadvantage
Data travel at great Speed	Require more cable than bus
No collision	A break in ring will bring it down
No terminator req.	Not as Common as bus.

Note :

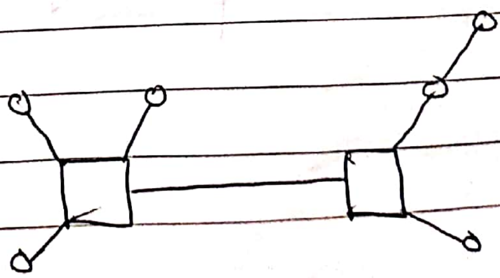
- For extra Safety double layer ring is Used.

Note :

Tree :



Hybrid :



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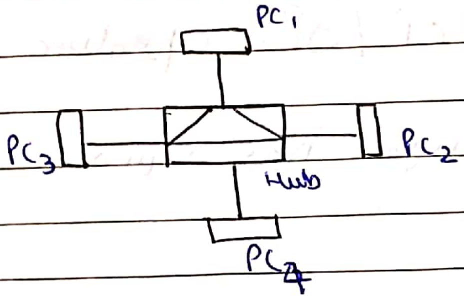
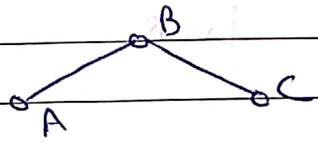
Star::



→ All Computers / devices Connect to Central device (hub or switch)

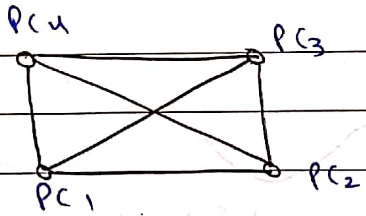
→ Each device requires a Single Cable.

→ Point to Point Connection between device & hub.



Mesh::

n = Nodes



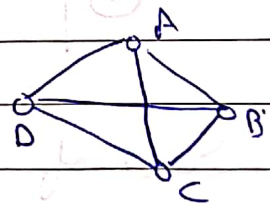
$\frac{n(n-1)}{2}$ = wires

$\frac{4(4-1)}{2}$ = 6 wires

→ Each nodes Connect to every other

→ Most often Used in WAN to interconnect LAN

Advantage → Improve Fault tolerance
→ Can Carry more data.



Disadvantage → Expensive, Difficult to Install.

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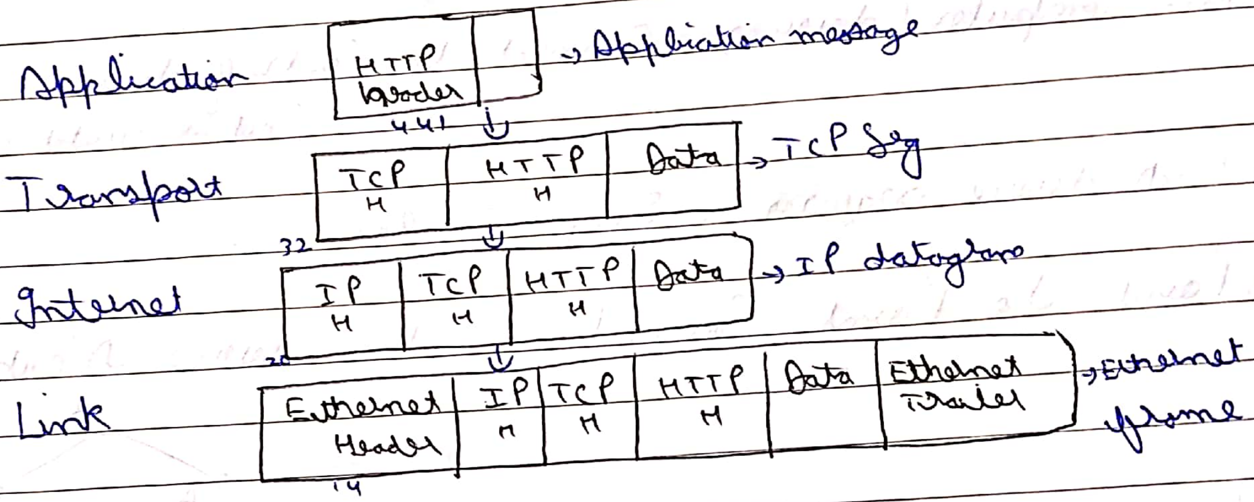
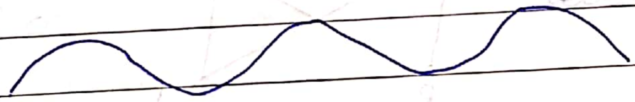
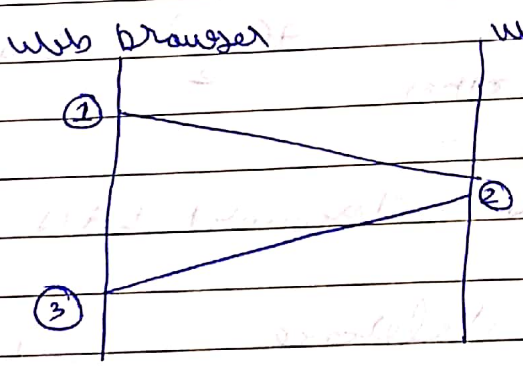
TCP / IP - operation

Assume - " Web browser has requested a webpage from Server



 = Signal



Size - 97 B

→ Size of the original webpage is 97 B, but after adding the header the total size become quiet larger than the initial webpage.

14	20	32	441	97

$\underbrace{14 + 20 + 32 + 441 + 97}_{604 \text{ Bytes}}$

$$\text{Efficiency} = \frac{97}{604} \times 100 = 16\%$$

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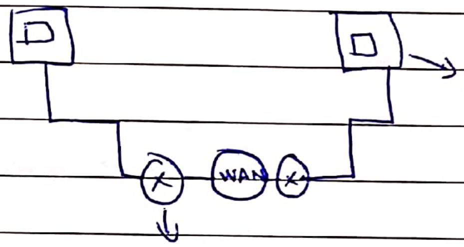
* DTE and DCE Interface ;

Data Terminal Equipment (DTE)	Data Communication Equipment (DCE)
→ Device that ends Communication line i.e Computer	Device provides path for Communication i.e modem



→ Interface Standard :

* Mechanical - Wiring and Connector
* Electrical - Signal
* Functional - Protocols for Signals

{ 25 Pins Set }

organization for Standard :

EIA = Electronic Industries Association
ITU-T = International Telecomm^n Union - Telecom Std.

RS-232 (Serial Port) = Recommended Std by EIA.

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* Mechanical Std :-

Male → DB-25 Connector (DTE)

Female → DB-25 Connector (DCE)

* Electrical Std :-

NRZ-L encoding (in Digital Sig)

1 → -3V to -15V
0 → +3V to +15V

* Functional Std :-

25 Pins Set :-

→ Most of the Pins for Controlling the Data
→ Pin-2 ⇒ Transmission of Data → DTE
→ Pin-3 ⇒ Receiving the Data → DCE

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→ Data and Signals (Analog and Digital)

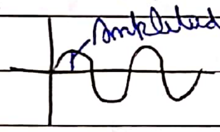
→ To be transmitted, data must be transformed to electromagnetic Signals.

Data

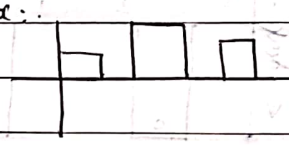
- Analog
  - → Continuous
- Digital
  - → Discrete.

Signal

- Analog
  - Ex:
  - →



- Digital
  - Ex:



→ Periodic

$S(t+T) = S(t)$

→ Aperiodic

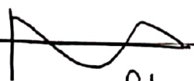
→ $f = \frac{1}{T}$

(Hz) T (seconds)

→ "Freq is rate of change with respect to time"

Phase

Phase = 0°



Phase = 90°

Wavelength = Propagation Speed × Period = $\boxed{\lambda = \frac{c}{f}}$

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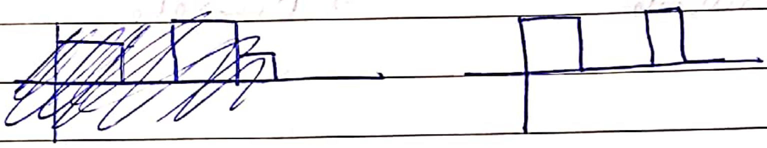
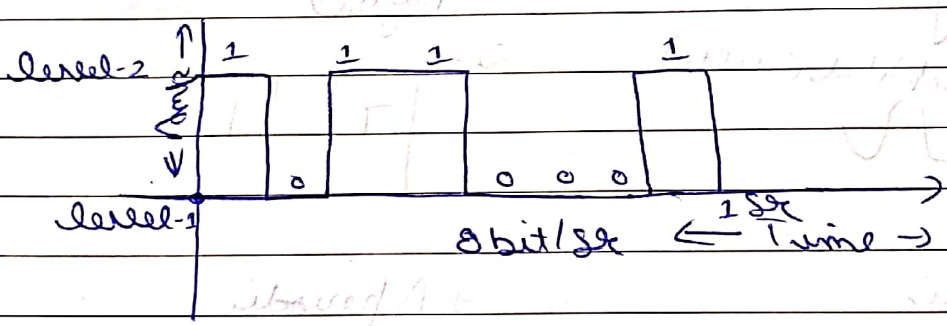
→ Composite Signal :
It is made up of many Simple Sine waves.

Bandwidth :
Range of frequencies contained in a Composite Sig.

→ Digital Signals : These are the type of Signals that are represented in the form of 0 & 1.



Bit rate = It is used to describe digital signal

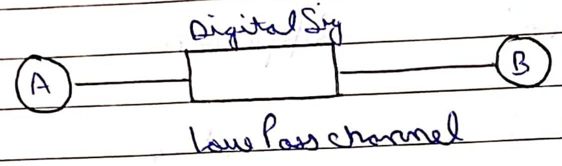


Bit length = It is the distance that one bit occupies in transmission medium

Bit length = Propagation Speed × bit duration

Transmission of Digital Signal :

Baseband Transmission :

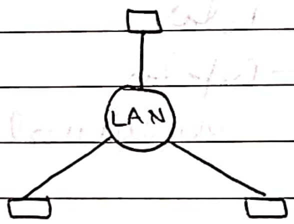


- It is the type of transmission that uses digital Signaling over a Single wire.

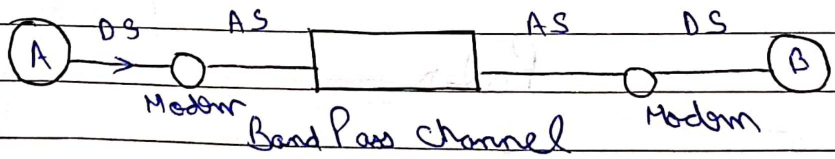
- Baseband transmission is bidirectional, allowing computers to both Send and receive data using a Single Cable.

- Digital Signal occupies the entire bandwidth of the network media to transmit a Single data Signal.

Ex :



Broadband Transmission :



- It uses analog Sig in the form of electromagnetic waves over multiple transmission freq.

- For transmission, digital Signal has to change in Analog Sig

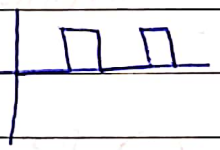
→ Digital Data to Digital Signal Conversion :

(Digital to Digital Conversion)

Data	Signal	Approach
Digital	Digital	Encoding
Analog	Digital	Encoding
Analog	Analog	Modulation
Digital	Analog	Modulation

→ "Signal Selection depends on the Situation and available bandwidth"

Line Coding (Digital to Digital Data Signal) → Polar, Bipolar, Multilevel

010101 → Line Coding → 

Polar → NRZ, RZ, Manchester, Diff manchester

1) NRZ (Non-Return to Zero)

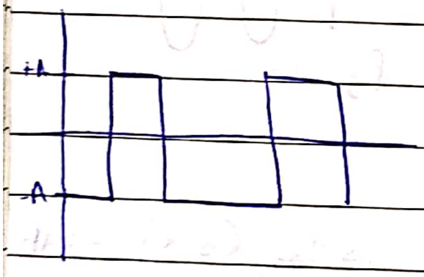
→ Voltage level is constant in bit interval.

NRZ

NRZ-L

1 = Low level
0 = High level

1 0 1 1 0

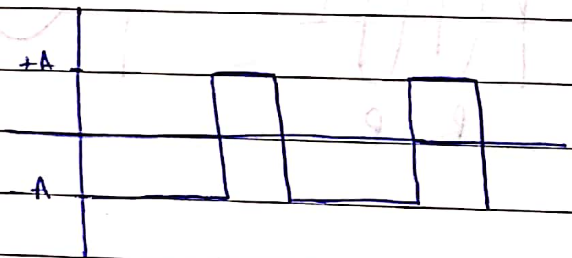


(Digital Sig)

NRZ-I (Inverted)

Transmission from one voltage to the other represents as = 1

1 0 1 1 0 1



In this form level changes with respect to the 1 that comes one after the other (Low → High)

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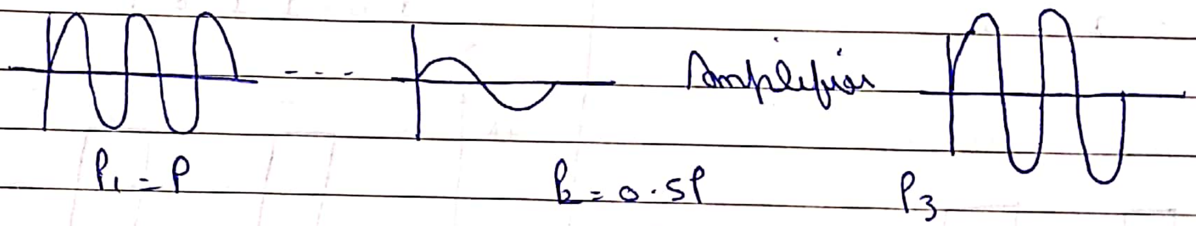
→ Transmission Impairment :-

" What is Sent, is not what is received.

3 Causes of Impairment
- → Attenuation
- → Distortion
- → Noise.

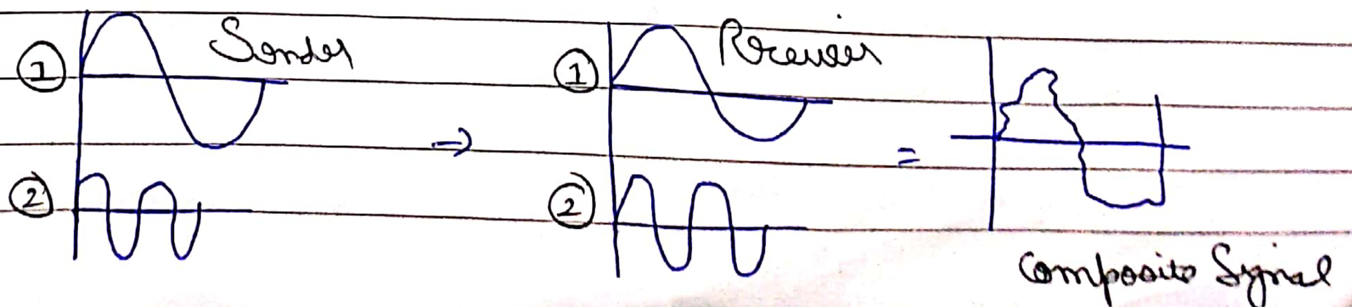
1) Attenuation : It means loss of energy.



$$dB = 10 \log_{10}\left(\frac{P_2}{P_1}\right) = 10 \log_{10}\left(\frac{0.5P}{P}\right) = 10 \log_{10}(0.5) = -3\,dB$$

P = Power at Particular point
dB = Decibel measures the relative strength of two sig or one sig at two point.

2) Distortion : In this Signal changes its form or shape distortion can occur in composite signal made up of different freq.



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→ We Can See that Composite Signal is distorted ~~[illegible]~~ it is because Signal ② at the receiver is delayed by Some time.

3) Noise: Several types of noise may interrupt with the Signal

→ Thermal Noise
→ Induced Noise.
→ Cross Talk
→ Impulse Noise.

→ Data Rate limit:

→ Data rate depends on 3 factor:

1) Bandwidth available.

2) Level of Signal we Use.

3) Quality of channel (level of noise).

→ To Calculate data rate we Use
→ Nyquist ~~[illegible]~~ bit rate (Noiseless channel)
→ Shanon Capacity (Noise)

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1) Nyquist Bit rate (For Noiseless)

Channel Capacity = Bit rate $= 2 \times B \times \log_2 L$

Bandwidth — No. of level

2) Shannon Capacity (For noisy)

$$C = B \cdot \log_2 (1 + SNR)$$

$$SNR = \frac{\text{avg Sig Power}}{\text{avg Noise Power}}$$

$$\boxed{SNR_{dB} = 10 \log_{10} SNR} \quad - (1)$$

Case 1

If $\boxed{S.N.R = 0}$ (Very Noisy Channel)

$C = B \cdot \log_2 (1+0) = 0$ (No data is transmitted)

Ex:

$B = 4\ KHz$

Signal to noise ratio - 20 dB ⇒ To be putte in (1)

$C = 4000 \times \log_2 (1 + 100) = 26.6$ Kb per Sec

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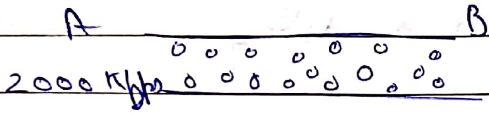
DCN = Performance of Network.

- Bandwidth = Range of frequencies a channel can pass. (Hz or bps) (1)

- Throughput = Amount of data successfully delivered to destination. (2)



100 kbps
↓
(Throughput)

- Delay = Time to transmit data from source to destⁿ. (3)

Unit = Seconds.

- 4 (Four) components.

- Transmission delay - Time to transmit data on the link

- Propagation delay - Time for signal element to propagate across the link.

- Processing delay - Time for device to process the data

Queuing delay = Time data spend waiting in a queue in a device.

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④ Latency = TD + PD + PD + QD

Jitter: (Problem)

If different packets of data encounter different delays.

⑤
Bandwidth - delay Product:
= Bandwidth × delay

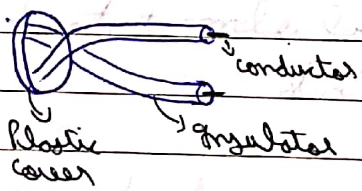
Transmission Media:

Physical Path between transmitter and receiver in a data communication

Guided (wired)

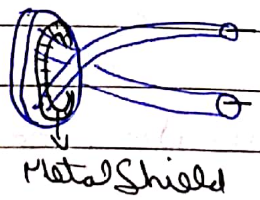
Unguided (wireless)
"Signal transmission through air".

① Twisted Pair:



UTP
↓
unshielded twisted pair

Cat1 - Telephone
Cat2
Cat3
Cat4
Cat5
} LAN



STP
↓
Shielded Twisted Pair

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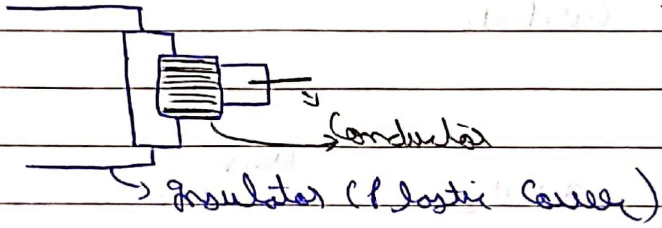
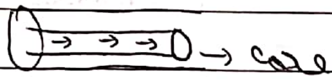
② Co-axial Cable :

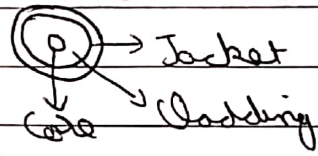


TV, LAN

Category	
RG-59	TV
RG-58	Thin Ethernet
RG-11	Thick Ethernet

③ Optical fibre :





→ Total Internal Reflection.

→ Multimode. XXXXX

→ Single mode → → → →

→ Multimode graded index

Optical fibre type

Type	Core (um)	Cladding (um)	Mode
50/125	50	125	Multimode G.g
100/125	100	125	"
7/125	7	125	Single mode

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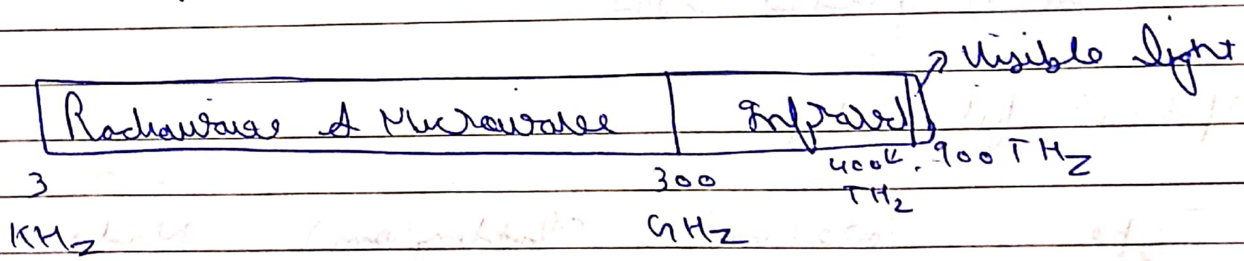
Comparison

Medium	Cost	B.W Data Rate	Attenuation	Security
UTP	Low	3 MHz	High 2-10 Km	Low
Co-axial	Moderate	350 MHz	Moderate 1-10 Km	Low
optical fibre	High	2 GHz	Low 10-100 Km	High

Unguided Media

"Transport electromagnetic waves without using physical conductor".

- Electromagnetic Spe for Wireless :



- Transmission + reception ⇒ Antennas

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↳ Ways :

1) Ground Propagation : Radio waves travels through lowest portion of atmosphere hugging the earth.

2) Sky Propagation : Higher freq. radio waves radiate upward in to the ionosphere where they are ~~S~~ reflected back to earth.

3) Line of Sight Propagation : Very high freq. Signals are transmitted in St line direction from antenna to antenna.

↳ Review of Error Detection and Correction

Why error occurs → Attenuation → Noise
→ Distortion → Interference during trans^n.

Types

1) Single Bit error 1011 → 0011

2) Burst error 10101101 → 10110001

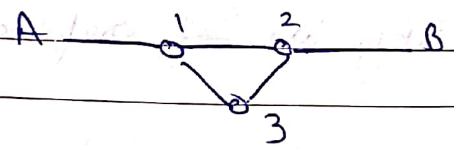
Technique

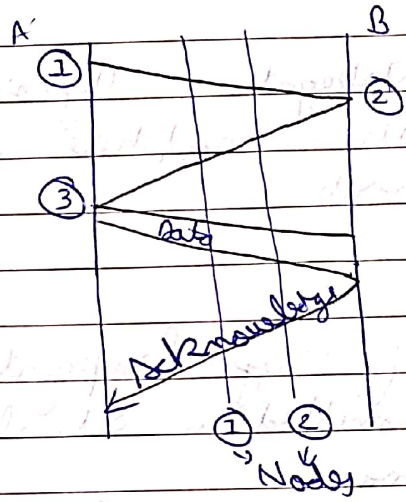
↳ Parity check (even or odd)
↳ Checksum
↳ Cyclic redundancy check

Switching Technique:

1) Circuit Switching: (Dedicated links req.)

→ To establish end-to-end connection **before** any transfer of data



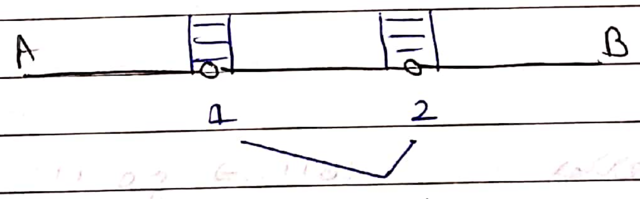


Possible Paths

→ A → 1 → 2 → B
→ A → 1 → 3 → B

Steps
→ Establish.
→ Transfer Data.
→ Disconnect.

2) Message switching:



→ In message switching the data we want to deliver to B firstly have to be stored at 1 and after that at 2, which means they must have significant space for the whole message to be transferred to B.

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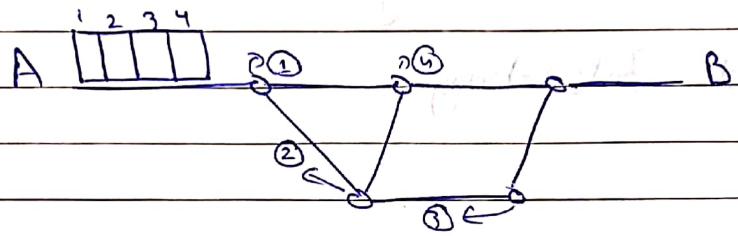
3) Pocket Switching :

→ In Pocket switching the main data can be divided into pockets so that they can be delivers easily through the nodes to (B)



Circuit Switching

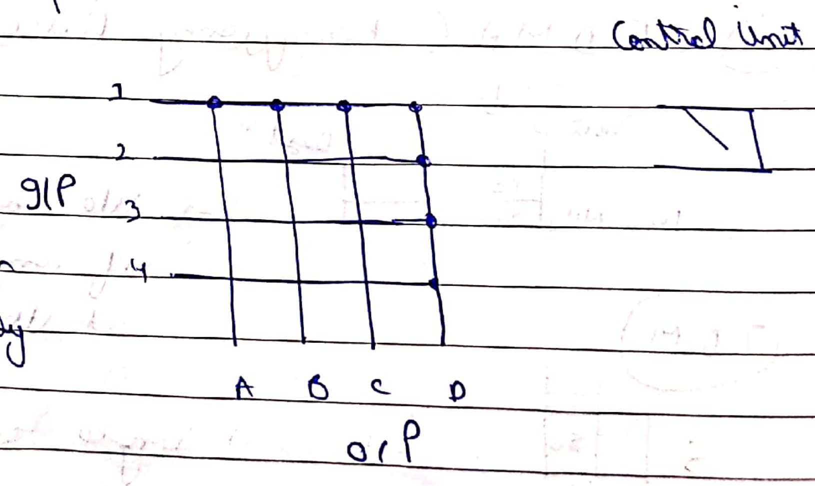
Space Division Switching | Time Division switching | Comb^n of (SDS + TDS)

Space Division Switching : Path in circuit are separated with each other spatially.

Ex: → Crossbar Switching
→ Multistage switching

1) Crossbar switches

→ It is metallic cross pt or Semi cond gate that can be enabled or disabled by Control unit



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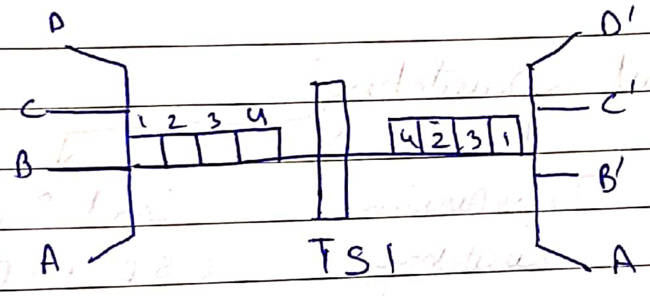
Disadvantage :-

→ It is very ineffective that remaining points are unused, So they will remain idle.

→ Time Division switching :

1) Time Slot Interchange.

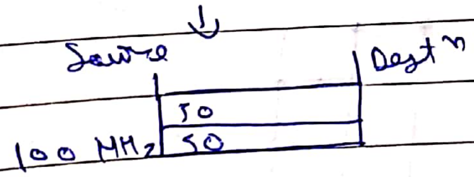
2) TDM Bus



(1)

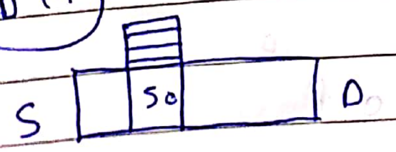
TSI allows us to interchange the delivery of packets when received as input.

(2) FDMA (Frequency Division Multiplexing)



⇒ We cannot vary the freq if engage, we can only vary at the time of disconnection

TDM



⇒ Single Source but freq can be vary within 3 diff time slot.

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Packet Switching Technique

Virtual Circuit — Datagram

↳ Packet Switching:

↳ Message is broken into the series of small packets

↳ Every packet contains some control infⁿ in its header. (for routing)

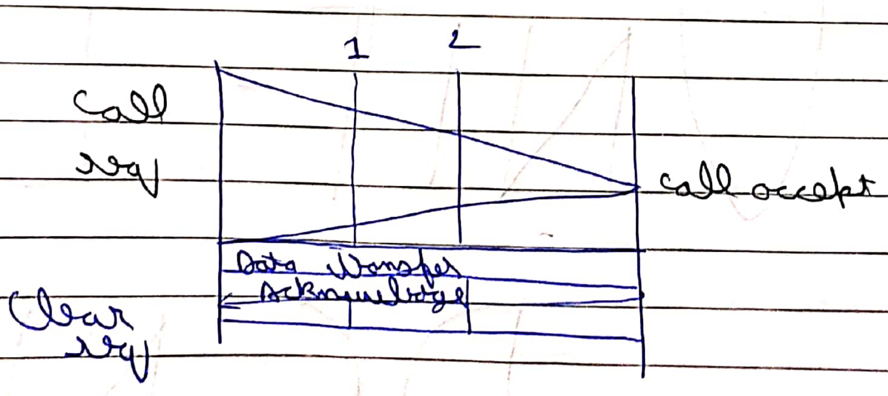
↳ Virtual Circuit Approach:

* A preplanned route is established before any packets are sent

↳ Call req & Call accept packets are used to establish the connection.

↳ Route is fixed for the duration of logical connection

↳ A Clear req packet is used by one to terminate.



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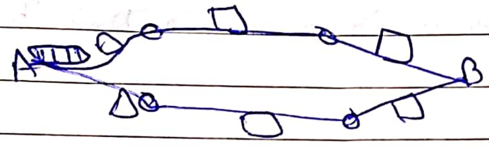
ii) Datagram approach :-

* Each packet is treated independently with no reference to packets that have gone before

* Every intermediate node has to take routing routing decision



VCA	DG
Nodes = No routing decision	Nodes = Routing decision

UNIT-1

DCN